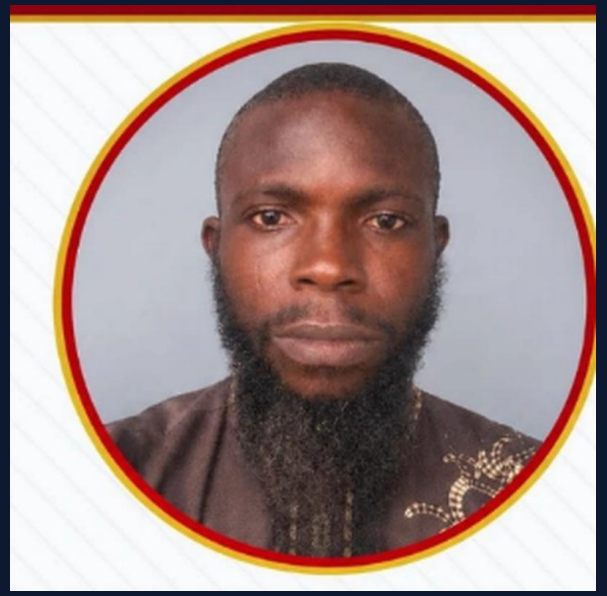
PERSONAL NOTES API

Backend Engineering Project Demonstration

REST API for a personal knowledge base



PROJECT INTERN

Presented by

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Backend Intern • BeTechified

Project Overview

01

The goal

A REST API for a personal second-brain application

02

Technology

Node.js, Express, MongoDB Atlas and Mongoose

03

Core capability

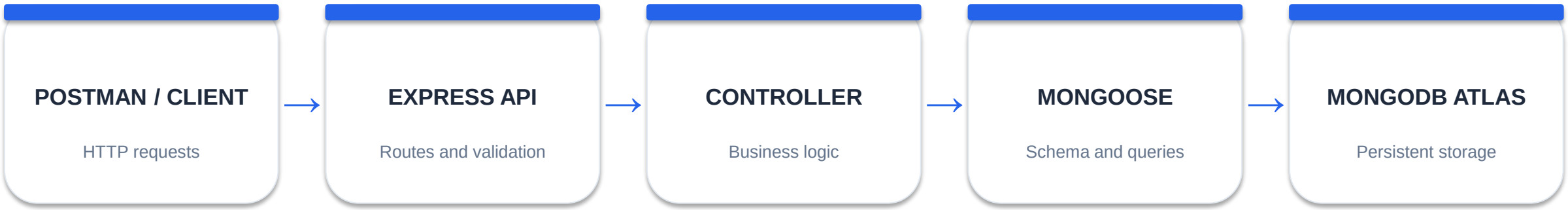
Validated CRUD operations

04

Advanced queries

Search, category, pagination and sorting

Architecture & Technology Stack



Node.js • Express • MongoDB Atlas • Mongoose • Joi • dotenv • Render

Data Model

Note document

- title: String • required
- content: String • required
- category: String • default General
- tags: String[]
- createdAt: Date • automatic
- updatedAt: Date • automatic

MongoDB text index

{ title: "text", content: "text" }

Indexes support full-text keyword search across note titles and content.

One note represents a single item of personal knowledge, such as a meeting reminder, idea or reading list.

API Query Features & Demonstration

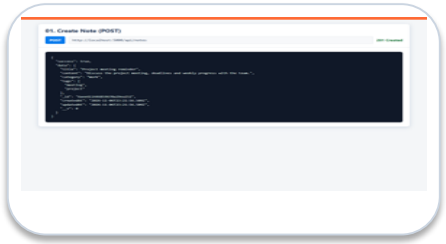
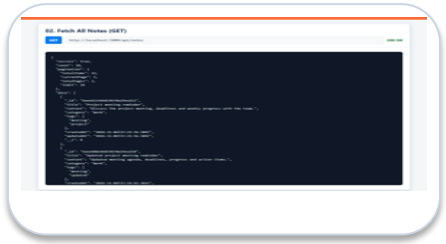
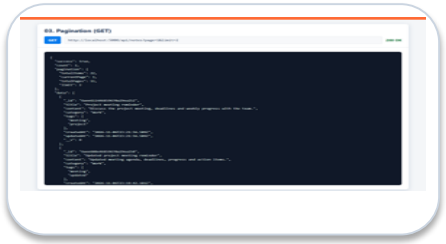
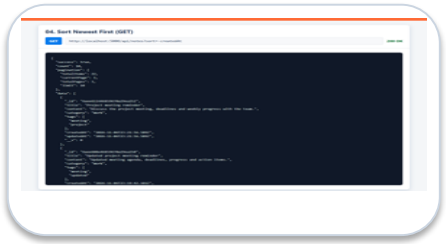
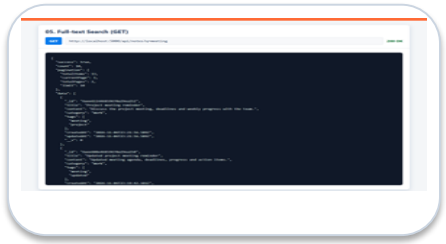
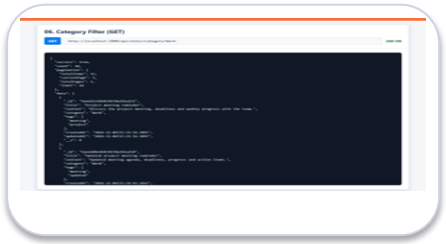
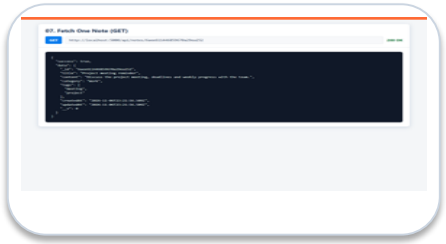
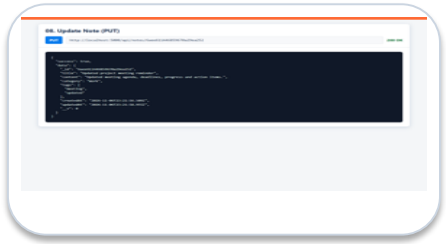
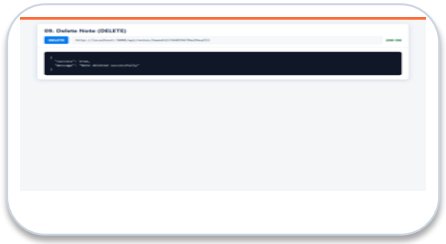
CRUD	POST • GET • PUT • DELETE
PAGINATION	page=1 & limit=10
SORTING	sort=-createdAt
TEXT SEARCH	q=meeting or search=meeting
CATEGORY	category=Work

Live API demonstration

1. Create five notes
2. Fetch a list and individual note
3. Update and delete evidence note
4. Combine page, sort, search and category
5. Validate success codes and JSON responses

https://personal-notes-api-bavd.onrender.com/api/notes

Challenges, Learnings & Test Evidence

Challenges & learnings

- MongoDB connection reliability
- Text-index configuration
- Handling validation consistently
- Combining multiple query filters
- Deploying environment variables
- Producing repeatable API evidence

Deployment & Conclusion

A production-ready personal knowledge API

LIVE DEPLOYMENT

Personal Notes API

personal-notes-api-bavd.onrender.com

MongoDB Atlas connected • Public Render service • Verified HTTP 200

READY FOR FINAL SUBMISSION

Delivered

- ✓ Validated CRUD API
- ✓ Advanced query features
- ✓ Automated evidence
- ✓ Complete README
- ✓ Public deployment
- ✓ Presentation material

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